

Austin Zhai

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EDUCATION

University of British Columbia

Vancouver, BC

Bachelor of Applied Science in Computer Engineering

Graduation Date: April 2029

- Cumulative Average: 83.7% | GPA: 3.90/4.33 | Honors: Dean's List | Relevant Coursework: Circuit Analysis and Electrostatics: A-

EXPERIENCE

Operational Technology Engineering Intern

Calgary, AB

Galaxy Instrumentation and Controls Inc

May 2026 – Present

- Audit CHEP OT infrastructure across 750+ global manufacturing sites at ~25 sites per week by executing pre and post migration network verification supporting the Perfect Plant initiative across FMS servers, PLCs, HMIs, and ADI devices
- Validate connectivity across ~7 Allen-Bradley PLCs per site by configuring Ethernet communication drivers in RSLinx Classic Lite, verifying module and port configurations, and flagging communication failures for post-migration review
- Confirm ADI system health with heartbeat intervals under 300ms by remotely connecting via Radmin, querying production history and alarm event tables in SSMS, and verifying data transfer speeds
- Cut manual audit setup time by ~15% by building a Python script integrated with Microsoft Power Automate to automate file and folder navigation across multi-site documentation workflows

IoT Cyber Defense Extern

Remote

Hydroficient

April 2026 – Present

- Eliminated unauthorized data injection across 3 simulated attack classes by designing and stress-testing a 5-layer security pipeline using TLS, mTLS, HMAC signing, timestamp validation, and sequence counters across a 500-node IoT network
- Detect anomalous sensor readings in real time by implementing a Python/MQTT data acquisition pipeline and deploying an Isolation Forest model integrated into a Streamlit monitoring dashboard

Bilingual Data Analyst

Remote

TELUS Digital

March 2026 – Present

- Refined AI-powered mapping models by evaluating and querying 80+ weekly bilingual queries in French and English, improving POI validity and search relevance for millions of global users.

Operations Team Member

Vancouver, BC

UBC Sailbot

September 2025 – April 2026

- Grew social media engagement by ~80% and secured 18 active sponsors by independently managing a brand campaign across content planning, graphic design, and video production

Founder and Operator, Independent Haircutting Business

Calgary, AB

Self-employed

August 2024 – December 2025

- Generated \$7,000+ revenue at ~85% retention by building a 50+ client haircutting business through Excel-based referral system

PROJECTS

Servo Sonar Radar

[Github](#)

Arduino, C++, Processing

May 2026

- Built a real-time object detection system with a 180° sweep range and 3-mode state machine (auto-sweep, pause-on-detect, joystick control) by mounting an HC-SR04 ultrasonic sensor on a servo motor with millis()-based non-blocking timing
- Visualized live sonar data on a custom radar display rendering angle, distance, and proximity alerts in real time by implementing a Serial communication pipeline between Arduino and Processing at 9600 baud

Smart Gate Access System

[Github](#)

ESP32, C++, RFID, I2C, SPI

May 2026

- Built a hardware access control system combining RFID card scanning and 4x4 keypad authentication to actuate a servo-controlled gate, integrating SPI (RFID module), I2C (LCD display), and GPIO (LED/buzzer) peripherals on a single ESP32
- Implemented dual-factor authentication logic in C++ with real-time status feedback on a 16x2 I2C LCD and hardware alerts for granted and denied access events, including all hand-soldered connections

SKILLS & INTERESTS

Languages: Python, C, C++, Arduino, Javascript/Typescript

Hardware: Microcontrollers, Soldering, Breadboard Prototyping, I2C/SPI Protocols, Circuit Assembly

Technologies: Altium, KiCAD, Git/GitHub, Visual Studio Code, Arduino IDE, REST APIs, SQL Server, Power Automate

Interests: Personal Finance, Weight Training, Basketball/Sports, Piano